



Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI,
RANCHI

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GOC

MULTI CORRECT ANSWER TYPE QUESTIONS:

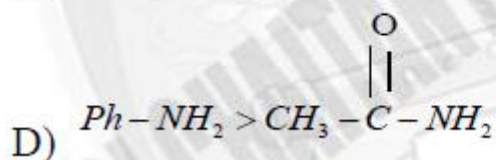
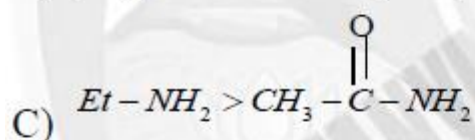
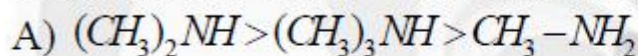
1.

which of the following is more acidic than ortho nitrophenol

- A) paranitrophenol B) Metanitrophenol
C) 2,4 dintrophenol D) 2,6 dintrophenol

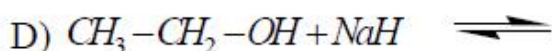
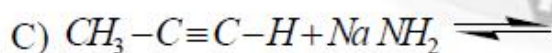
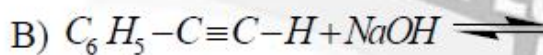
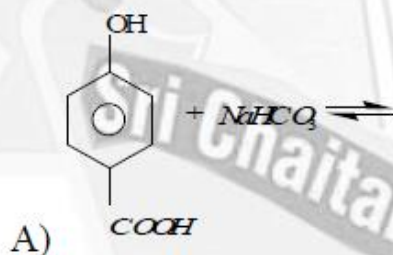
2.

Correct order of the basic strength among the following is (are)



3.

Different reactions are given below in which case($K_{eq} > 1$) equilibrium shifts forward direction



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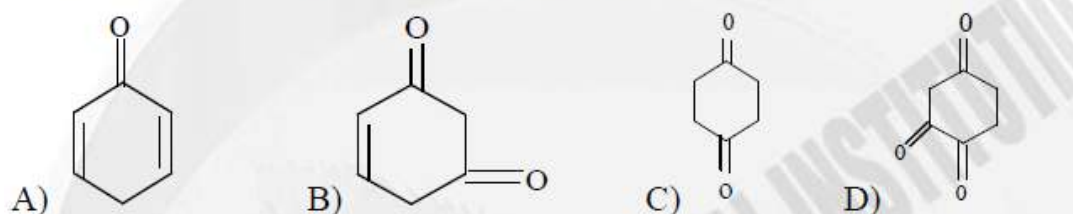
4.

For which of the following compounds ortho effect influences acidic or basic strength of the following

- A) orthonitro benzoic acid B) orthomethyl phenol
C) orthomethyl aniline D) orthochloro benzyl alcohol

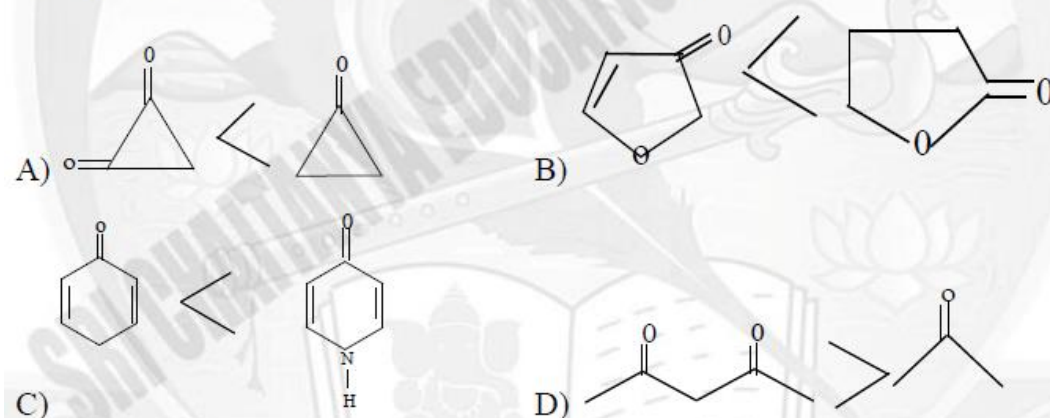
5.

Which of the following can exist in equilibrium with benzenoid form



6.

Incorrect order of enol content in ethanol solvent



7.

Which of the following statement is/are correct?

- A) The energy of resonance hybrid is always less than that of any resonating structure.
B) The resonance energy is the difference between the enthalpies of formation of the molecule and the resonating structure having maximum energy.
C) The resonance structures are hypothetical structure and they do not represent any real molecule.
D) In delocalized structure of benzene the π -charge cloud is spread equally above below the plane of molecule.

8.

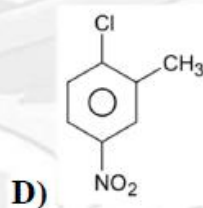
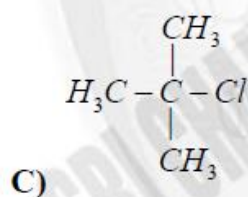
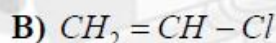
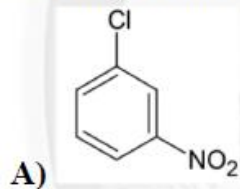
Which of the following is/are acceptable resonating structures of Buta-1, 2, 3-triene.



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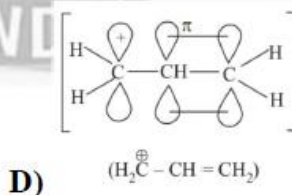
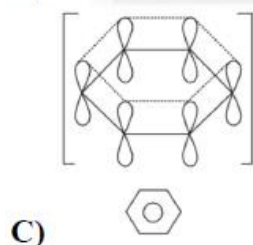
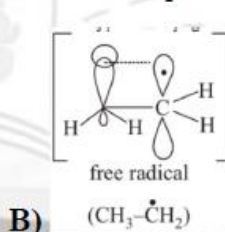
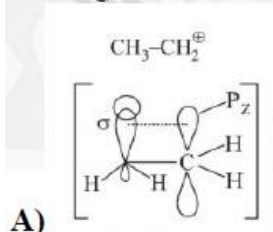
9.

In which of the following compound(s) the $C-Cl$ bond is shorter than that in C_2H_5Cl ?



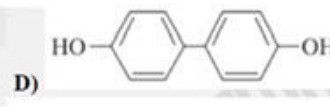
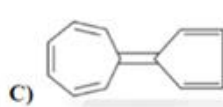
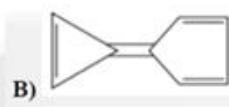
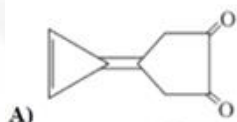
10.

Among the following orbital interactions which represents hyper-conjugation interaction?



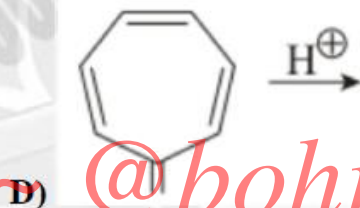
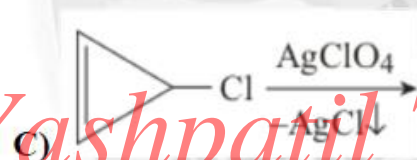
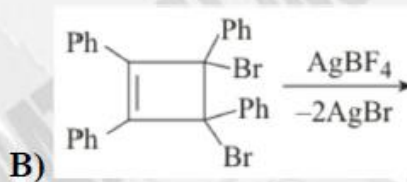
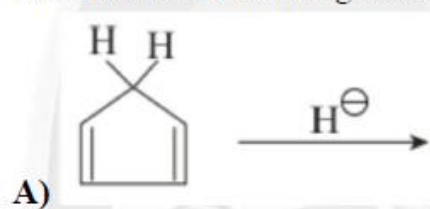
11.

Which has non-zero dipole moment?



12.

Which of the following reactions give an aromatic product?



PARAGRAPH QUESTIONS:

PARAGRAPH FOR QUESTIONS 13 AND 14:

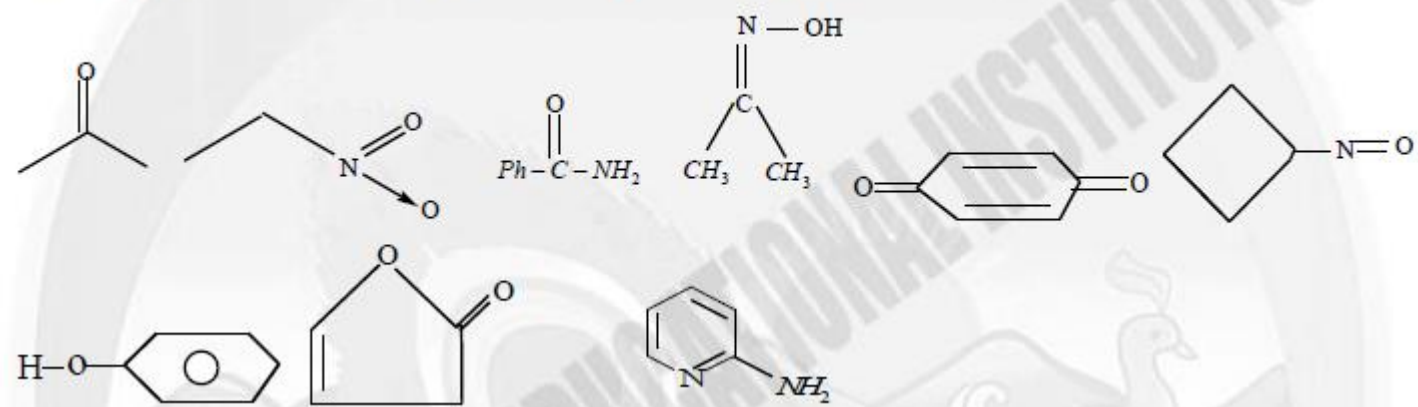
In the dyad tautomerism hydrogen atom oscillates between two polyvalent atoms linked together. In triad tautomerism hydrogen atoms travel from first to third atom in chain.

13.

Among the isomeric nitroalkanes of formula $C_4H_9NO_2$ how many isomers exhibit tautomerism (excluding stereo)?

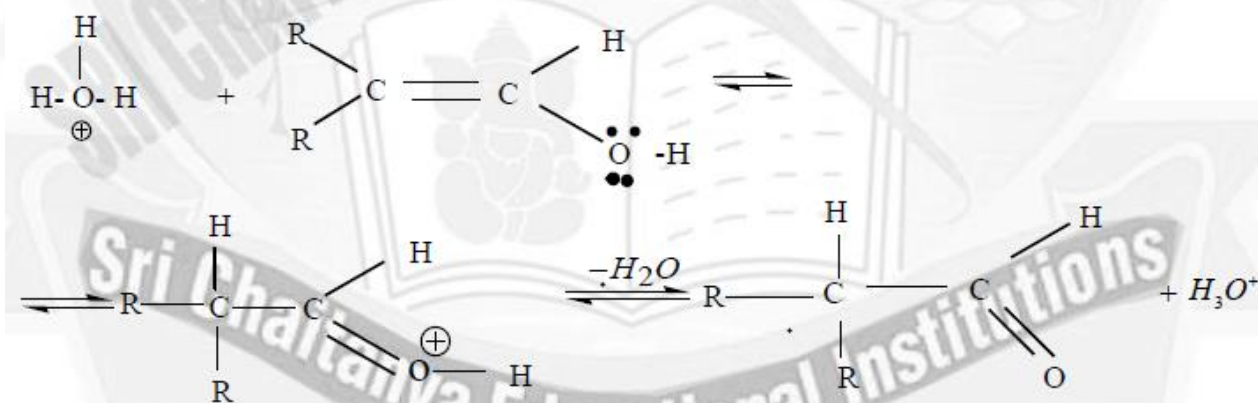
14.

How many of the following molecules exhibit triad tautomerism?



PARAGRAPH FOR QUESTIONS 15 AND 16:

Acid catalysed tautomerism can be given by the following mechanism in the presence of acid. The mechanism involves formation of enol.



If D^+ / D_2O is used instead of H^+ / H_2O Deuterium exchange reactions taking place

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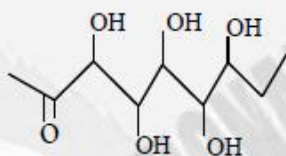
15.

The number of hydrogens exchanged with deuterium atoms in the following compounds after prolonged time (maximum) are x then $\frac{x}{2}$ can be



16.

The number of enol forms for the following compound is / are (assume no ring chain isomerism) and consider all possible enol forms by the transfer of protons when H^+ / H_2O are used. (excluding stereo isomers) consider trace amounts also.

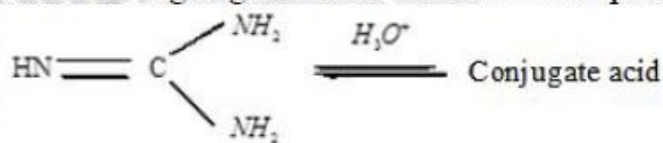


PARAGRAPH FOR QUESTIONS 17 AND 18:

Resonance can change the basic strength of amines in which lone pair electrons can participate in delocalization. And the stability of conjugate acid. The basic strength depends upon the donating ability of an atom containing lone pair of electrons

17.

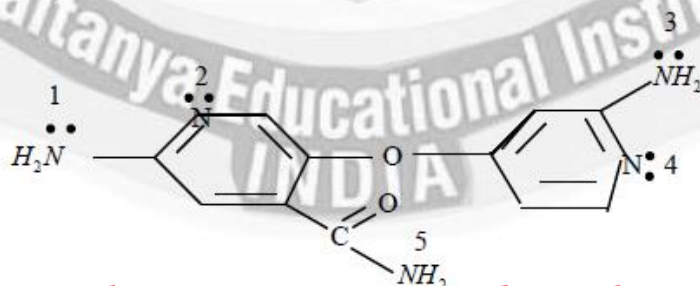
Guanidene is one of the strong organic base reacts with proton as follows



In the conjugate acid of guanidine how many electrons are involved in resonance

18.

In the following compound which numbered nitrogen atom has more tendency to donate lone pair of electrons to proton



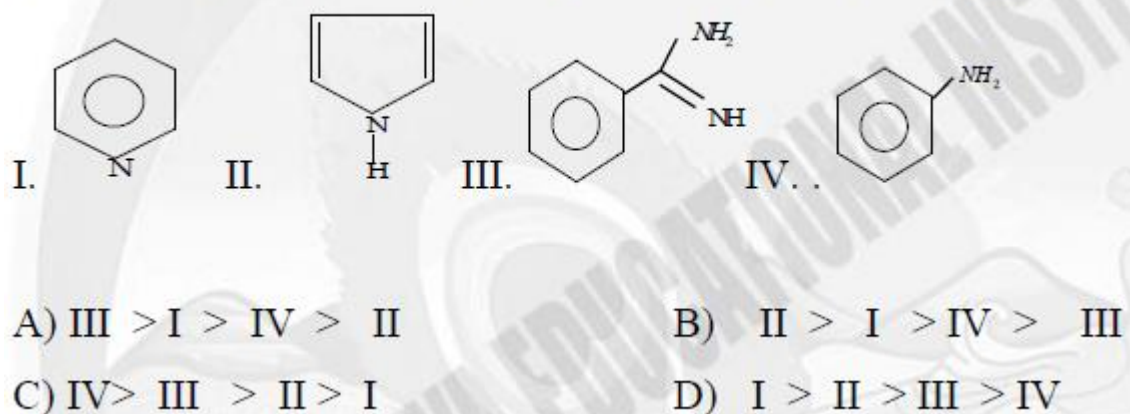
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PARAGRAPH FOR QUESTIONS 19 AND 20:

Acidic and basic strength depends upon several factors such as inductive effect, resonance hyper conjugation steric effect, solvent effect etc

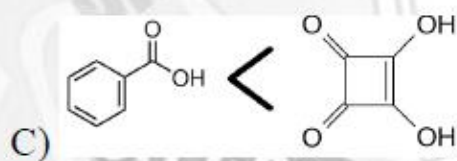
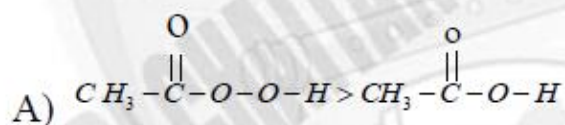
19.

Decreasing order of basic character of the following is



20.

Correct statement of the following regarding acidic strength of the following



PARAGRAPH FOR QUESTIONS 21 AND 22:

Acidic strength of protonic acids depends up on stability of conjugate base. Conjugate to base stability depends up on resonance inductive effect etc. when acid is strong its conjugate base is weak.

21.

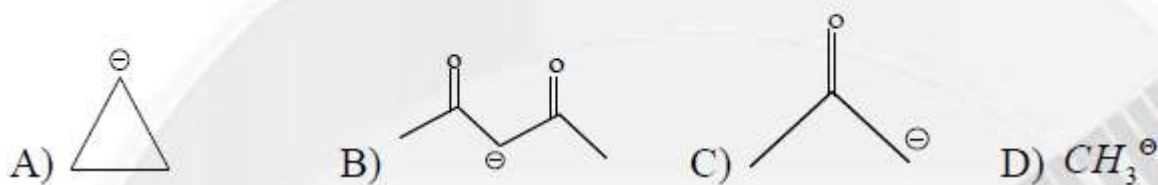
Strongest acid among the following is



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22.

Weakest base among the following is



NUMERICAL ANSWER TYPE QUESTIONS:

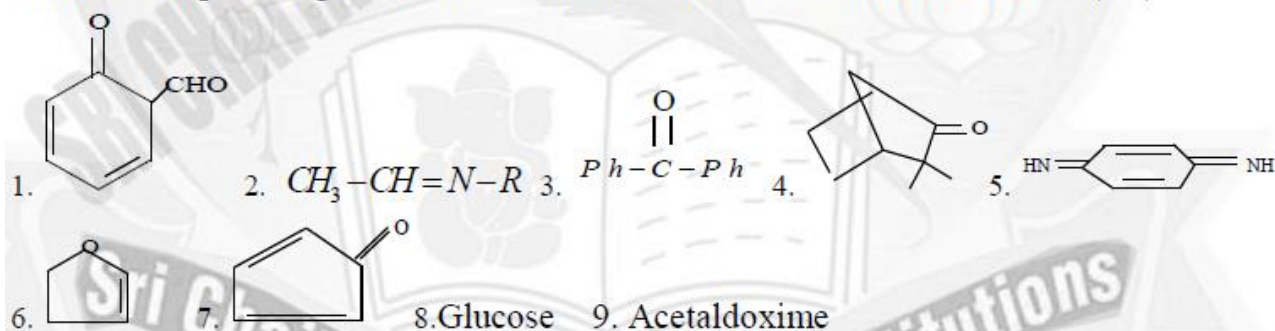
23.

How many of the following acidic compounds gives effervesces with NaHCO_3

1. $\text{CH}_3\text{-COOH}$ 2. Phenol 3. Picric acid . 4. 2,4-di nitro phenol 5. $\text{CH}_3\text{-SO}_3\text{H}$ 6. $\text{CH}_3\text{-CH}_2\text{-OH}$ 7. squaric acid

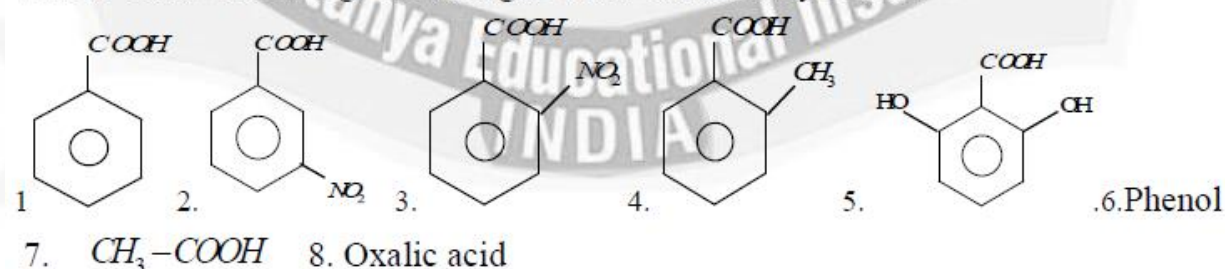
24.

Number compounds given below which can not exhibit tautomerism is (are)



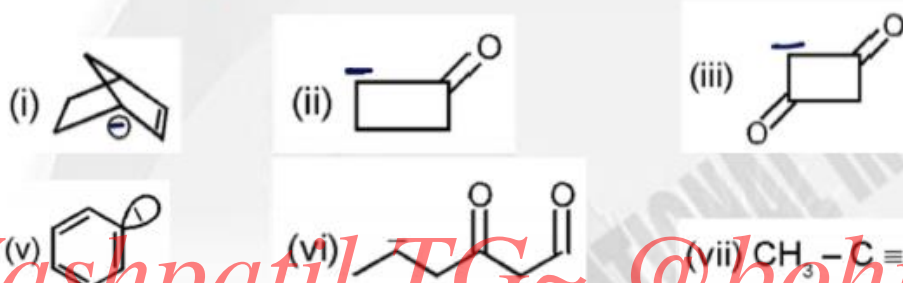
25.

Which of the following are stronger acids than salicylic acid



26.

In how many species the -ve charge (anion) which is not delocalized by resonance, are :



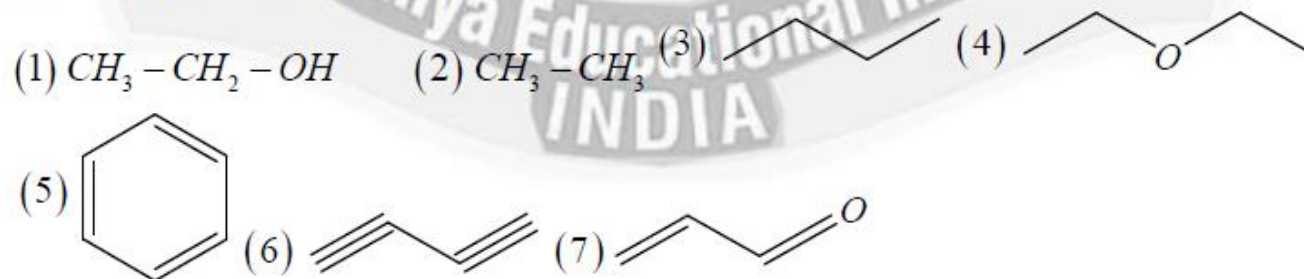
27.

Which of the following statements is (are) true about resonance.

- (a) Resonance is an intramolecular process.
- (b) Resonance involves delocalization of p electrons only
- (c) Resonance decreases potential energy of a molecule
- (d) Resonance is not the only way to increase molecular stability
- (e) The canonical structure explains all features of a molecule
- (f) The resonance hybrid explains all features of a molecule
- (g) Resonance hybrid is real and resonating structures are imaginary
- (h) Resonance hybrid is always more stable than all canonical structures

28.

How many can show inductive effect



29.

From the following compounds/ions :

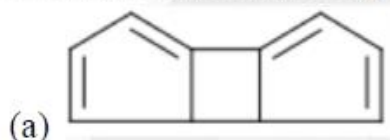


Identify value of "X".

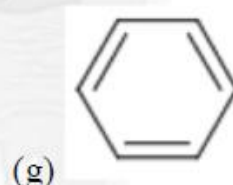
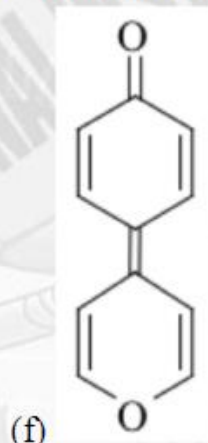
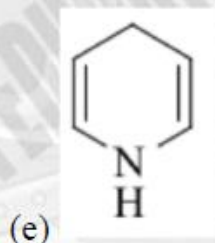
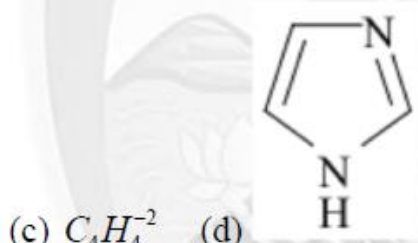
Where "X" is the total number of $E^{\oplus}$ / electrophiles.

30.

From the following compound



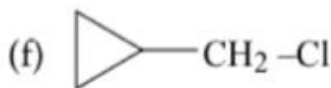
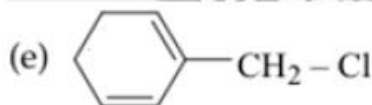
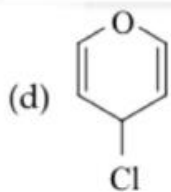
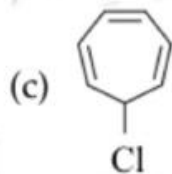
(b) $C_{14}H_{14}$ (use 2 six membered ring)



Identify the total number of aromatic compounds.

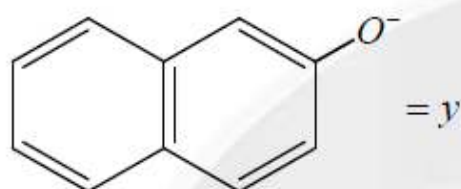
31.

How many products will be aromatic ion when $AgNO_3$ reacts with the following compounds?



32.

Find no. of resonating structure of = x

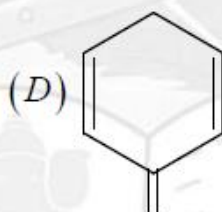
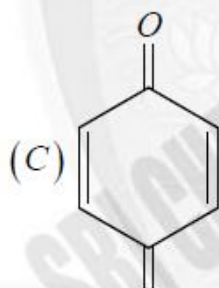
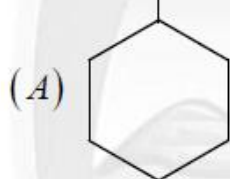


then $\frac{x+y}{2}$ is

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33.

The total number of contributing structures in A, B, C, D showing hyperconjugation (involving $C-H$ bonds) then find the total number of $A+B+C+D$ is

**KEY:**

1)	ACD	2)	CD	3)	ACD	4)	AC	5)	ABD
6)	ABC	7)	ACD	8)	ABC	9)	ABD	10)	AB
11)	ABCD	12)	ABCD	13)	3	14)	8	15)	6
16)	7	17)	6	18)	4	19)	A	20)	C
21)	C	22)	B	23)	5	24)	5	25)	3
26)	4	27)	7	28)	3	29)	4	30)	5
31)	3	32)	6	33)	8				